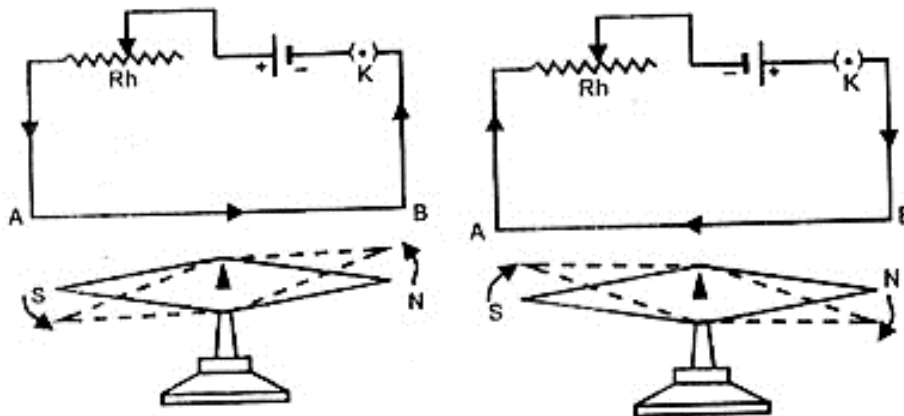


Oersted's Experiments: In 1819 a Danish scientist named Hans Christian Oersted discovered a close relationship between electricity and magnetism with this experiment. The new knowledge led to the study of electromagnetism.

According to the experiments, a conducting wire is kept along the length of a freely placed magnetic needle and a bit above it. If electric current is passed through the wire, it is found that magnetic needle is deflected from its original position. The magnetic needle tries to align itself at right angles to the conductor. But the needle does not align itself at right angles, because there is also the influence of earth's magnetism. As a result, due to the combined influence of magnetic field produced by the electric current and earth's magnetic field, the needle is deflected from the magnetic meridian and remain in equilibrium position.



If current is increased in the conductor, then the deflection of the needle also increases. If the direction of the current is reversed in the conductor, in that case the magnetic needle also deflects but in opposite direction to the earlier.

Again, if this experiment is performed keeping the wire below the needle, the needle also deflects. Keeping the direction of the current same, if the conductor is placed below the needle, then the direction of the rotation of the needle would be opposite if the conductor is placed above the needle. If the current is stopped in the conductor, then the needle returns to its original position.

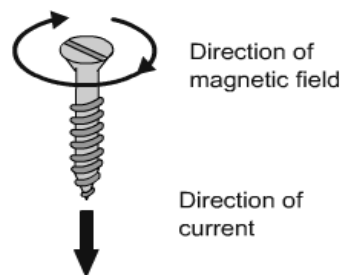
The following conclusions can be drawn from the Oersted's Experiment:

1. When current flows through a conductor magnetic field is produced around it.
2. Magnitude of the magnetic field depends on the strength of current.
3. The direction of magnetic field depends on the direction of the current flow.
4. Magnetic field exists as long as the flow of current exists in the wire.

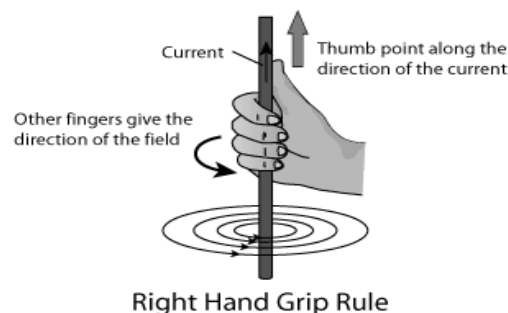
Direction of the magnetic field produced by electric current:

The direction of the magnetic field can be found using any of the following law:

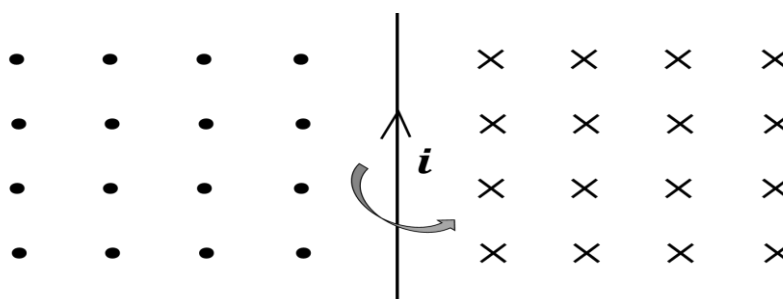
1. **Maxwell's cork screw law:** If a right handed screw is rotated along a current carrying conductor in the direction of the current, then the direction of the magnetic field will be in the direction in which the thumb rotates, i.e., the north pole of the magnetic needle will be deflected in this direction.



2. **Fleming's right hand rule:** If a current carrying conductor is grasped with the palm of the right hand so that the thumb point the direction of the current, then the tips of the other fingers will indicate the direction of the magnetic field.



Representation of Magnetic field in plane of paper: According to the Fleming's right hand rule, the dot sign ($\bullet$) is used to represent the magnetic field outward the plane or towards the reader and the cross sign ($\times$) represents inward direction of the field, as shown in figure below:



Magnetic Force: We know that a stationary charged particle produced electric field around it. Similarly we observed that a moving charge produced a magnetic field around it. Beside the electric Coulombs force, a moving charge exerts other force on another moving charge. This velocity dependent force on charges is known as magnetic force.

From experiment, it is found that the magnitude of the force (F) acting on a magnet charge in a magnetic field is –

- (a) directly proportional to the magnitude of the charge (q).
- (b) directly proportional to the velocity (v) of the charge.
- (c) directly proportional to the magnitude of the magnetic field (B)
- (d) directly proportional to the *sine* of the angle (θ) between the direction of the velocity of the charge and the direction of the magnetic field.

Thus, $F \propto qvB\sin\theta$

In S.I. unit the constant of proportionality is 1. So,

$$F = qvB\sin\theta$$

$$\vec{F} = q (\vec{v} \times \vec{B})$$

Magnetic force on the current carrying wire is –

$$F = ilB\sin\theta$$

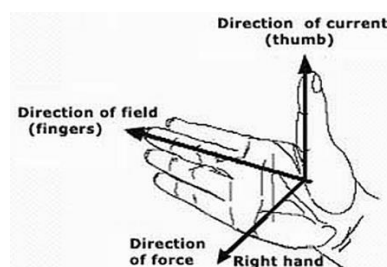
Where, i = Current flow through the conducting wire

l = Length of the wire

B = Magnetic field

θ = Angle between current flow direction and magnetic field.

Fleming's right hand rule – 1: When the right hand is stretched, if the fingers indicate the magnetic field and the thumb indicates the velocity of the charge or current flow direction, then outward normal drawn on the palm will indicate the direction of force for the positive charge or on the current carrying wire.



Some definitions:

- **Magnetic field:** The space or region around a magnet or a current carrying wire whereby the magnetic needle shows the deflection, that space or region is called magnetic field of that magnet or current carrying wire.
- **Magnetic flux:** Magnetic field is generally represented by magnetic field lines. If magnetic field is strong then lines are closer and density populated.

“Number of field lines passing through either a real or a fictitious plane in a magnetic field is called magnetic flux (ϕ)”

S.I. unit of magnetic flux is *Weber (Wb)* or NmA^{-1} .

1weber: Number of flux that changes per second in a coil for the production of 1volt potential difference in that coil is called 1Weber.

- **Magnitude of the Magnetic field (B):**

If a unit positive charge moving perpendicular to the magnetic field with unit velocity, then the force experienced by the charge is called the magnitude of the magnetic field.

$$B = \frac{F}{vq}$$

On the other hand, the number of magnetic field lines or magnetic flux incident normally on unit area of a real or fictitious plane is called flux density in that plane. It is called magnetic field or magnetic induction. i.e.,

$$B = \frac{\phi}{A}$$

S.I. unit of it is Wbm^{-2} or *Tesla(T)* or $NA^{-1}m^{-1}$. Also, $1Tesla = 10^4 Gauss$.

1Tesla: If 1Coulomb of charge is in motion perpendicular to direction of magnetica field with a velocity of $1ms^{-1}$ and experience a force of 1N, then the magnitude of that magnetic field is called 1Tesla (T).

Lorentz Force: The force that a charge experiences while moving simultaneously through the electric field and magnetic field is called Lorentz force.

If $\vec{B}$ is the magnetic field and $\vec{E}$ is the electric field, then the Lorentz force for the electron passing through both these fields is –

$$F = F_e + F_m$$

$$F = qE + qvB\sin\theta$$

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

Problem-1: An electron is moving with a speed of $5 \times 10^7 \text{ m/s}$ at right angle to a magnetic field of 5000G . (a) What is the magnetic force on the electron? (b) What is the radius of circle in which the electron moves?

Problem-2: When an electron moves with a velocity 10^5 ms^{-1} making an angle of 60° with a magnetic field of 0.50T then find the magnitude of the force acting on the electron.

Problem-3: When a proton moves with a velocity $2 \times 10^6 \text{ ms}^{-1}$ making an angle of 30° with a magnetic field at a certain place it experiences of force of $4.8 \times 10^{-15} \text{ N}$. Find the magnitude of the magnetic field.

Problem-4: When a 10cm long wire carrying 10A current is at an angle of 30° with a magnetic field then it experiences 5N force. Find the magnitude of the magnetic field.

Problem-5: At the equator, the earth's magnetic field is nearly horizontal, directed from the southern to northern hemisphere. Its magnitude is about 0.50G . Find the force on a 20m wire carrying a current of 30A parallel to the earth (a) from east to west (b) from north to south.

Problem-6: A straight horizontal segment of copper wire carries $i = 28\text{A}$. What are the magnitude and direction of the magnetic field needed of float the wire, that is, to balance its weight? Its linear mass density is 46.6g/m .

Problem-7: Find the force on each segment of the wire shown in figure below, if $B = 0.15\text{T}$. Assume that the current in the wire is 5A .

